Cellular changes and killing process

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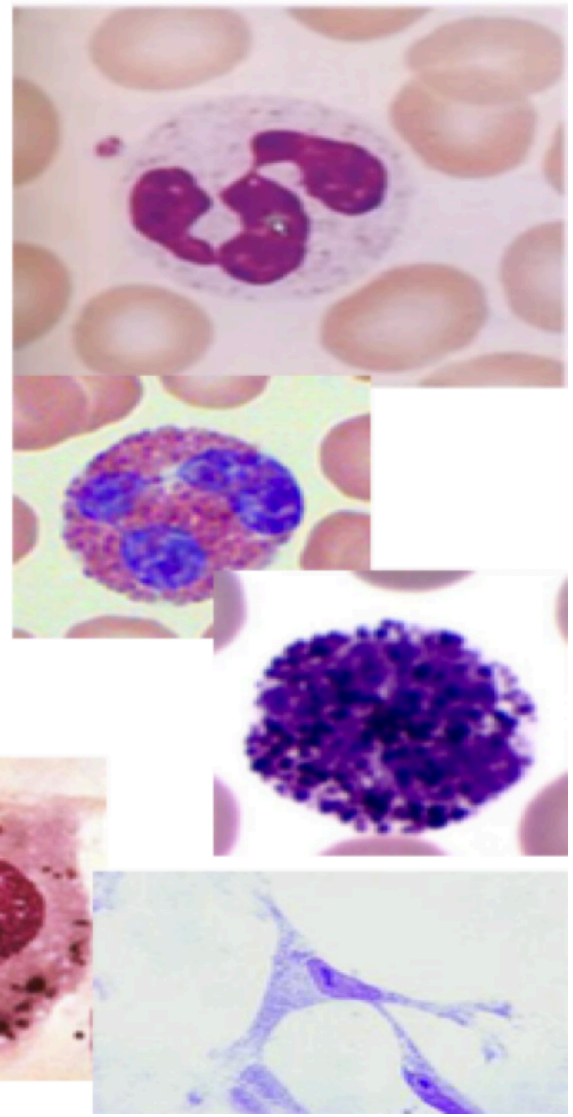
Learning outcome

- *Identify cellular changes that occur in acute inflammation*
- *State the process of phagocytosis*
- *Define and identify the morphologic pattern of acute inflammation*
- *Recognize acute phase response*

Cellular elements of inflammation

- ***Neutrophil granulocyte***
  - Major role in acute inflammation
  - Synonyms: Neutrophil, Polymorphonuclear, Leukocyte, PMN, PML, Granulocyte, „Poly", Polymorph
- ***Eosinophil granulocyte*** – parasites, worms
- ***Basophil granulocyte, mast cell*** – histamine, TNF storage
- ***Macrophages*** – phagocytosis, regulation of inflammation
- ***Endothelial cells*** – exudation, leukocyte migration
- ***Fibroblasts*** – regeneration
- ***Thrombocytes*** – PDGF, TGFβ, βFGF
- ***Lymphocytes***



II. CELLULAR EVENTS

- The cellular phase of inflammation consists of 2 processes:

1. Exudation of Leucocytes. (Influx to the site of Injury)

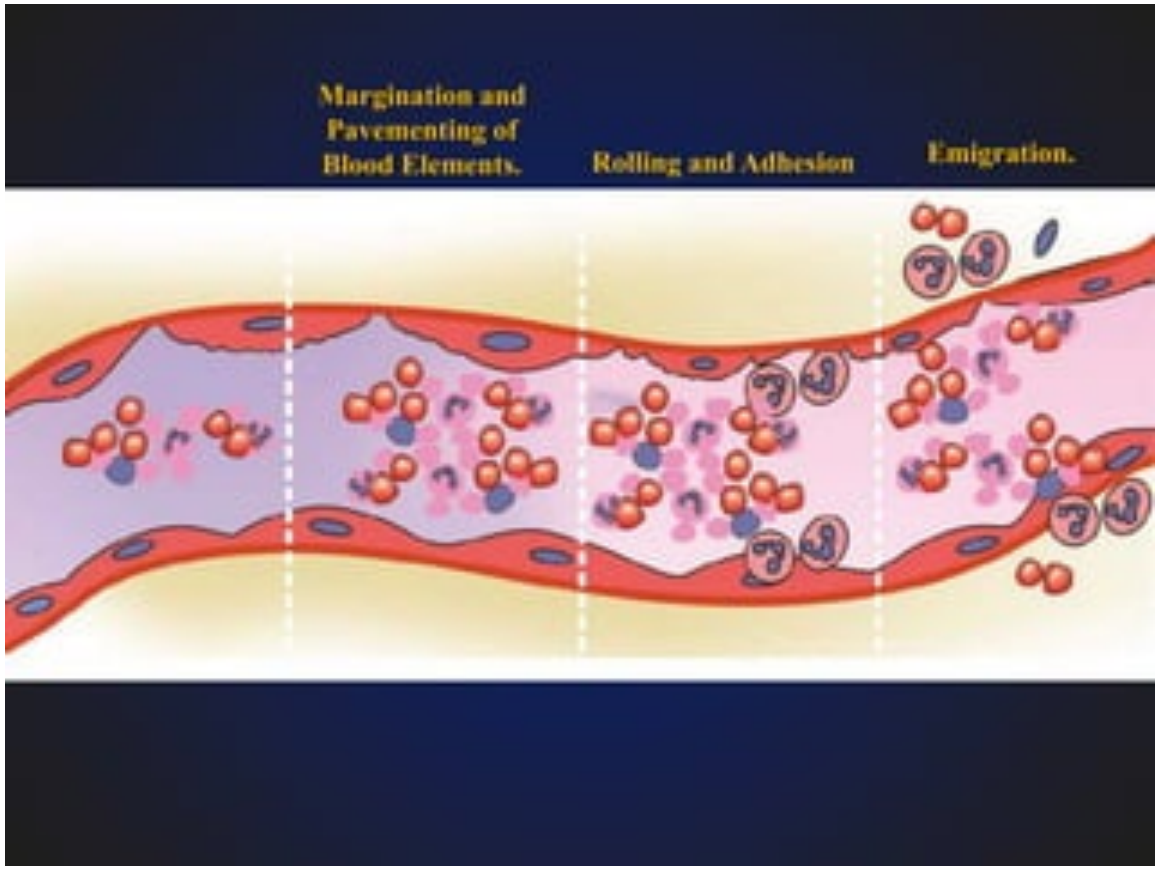
2. Phagocytosis. (Ingestion and Destruction of injurious Agents)

1. Exudation of Leucocytes.

- The escape of leucocytes from the lumen of microvasculature to the interstitial tissue is the most important feature of inflammatory response.
- In acute inflammation, ***polymorphonuclear neutrophils (PMNs) comprise the first line of body defense, followed later by monocytes and macrophages.***

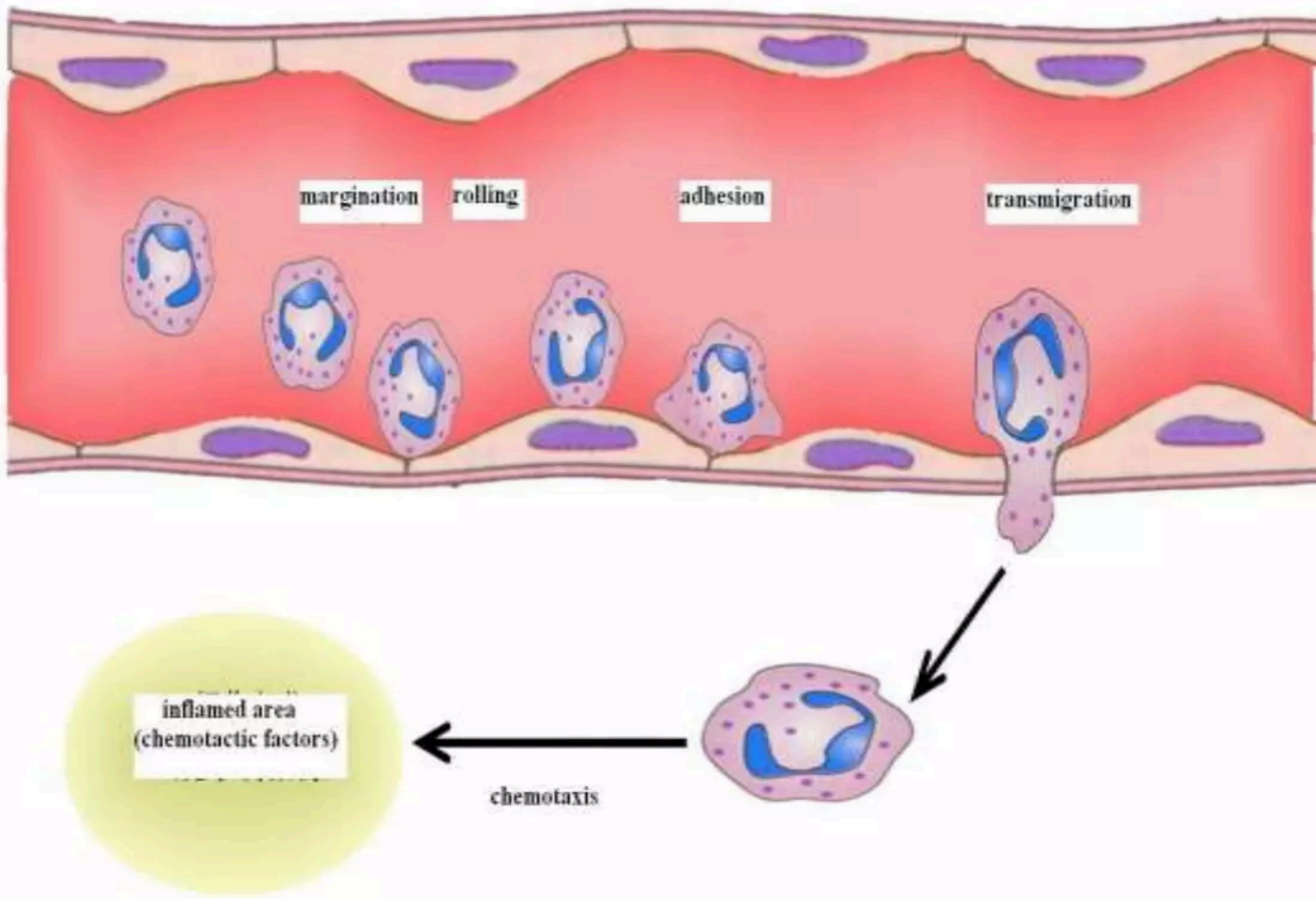
Extravasation;

This process occur in 4 ordered steps



Extravasation

- Margination → leukocytes assume peripheral portion of lumen
- Rolling & Adhesion → leukocytes tumble slowly along the endothelium and adhere transiently, detach and bind again.
- Diapedesis – Transmigration of leukocytes across the endothelium.
- Chemotaxis – leukocyte migration towards the site of injury (locomotion oriented along a chemical gradient)



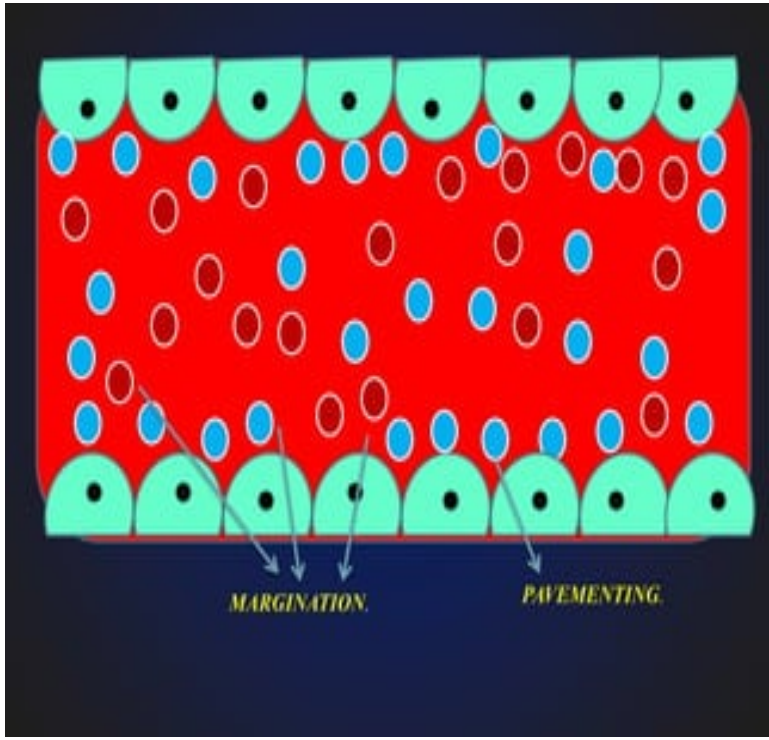
The process of leukocyte migration is multistep (Leukocytes Recruitment)

It is mediated and controlled by

- *adhesion molecules*
- *cytokines (chemokines).*

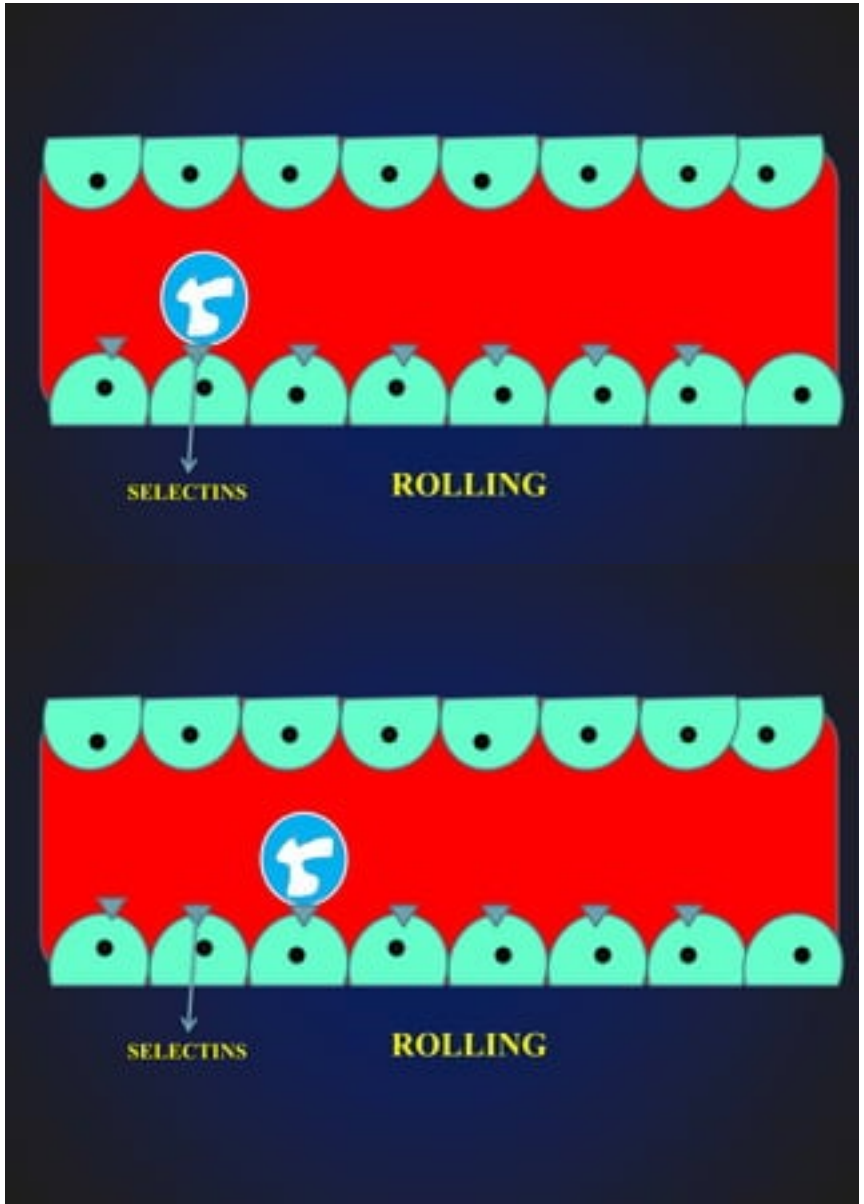
First step / Margination :

- *leukocyte accumulation at the periphery of vessels*
- *The white cells sense signals from the endothelium, and then start rolling & adhering to the endothelium via adhesion molecules, the expression of which is induced / enhanced by cytokines e.g. tumour necrosis factor (TNF), IL-1 and chemokines secreted by tissue macrophages, mast cells and endothelial cells.*

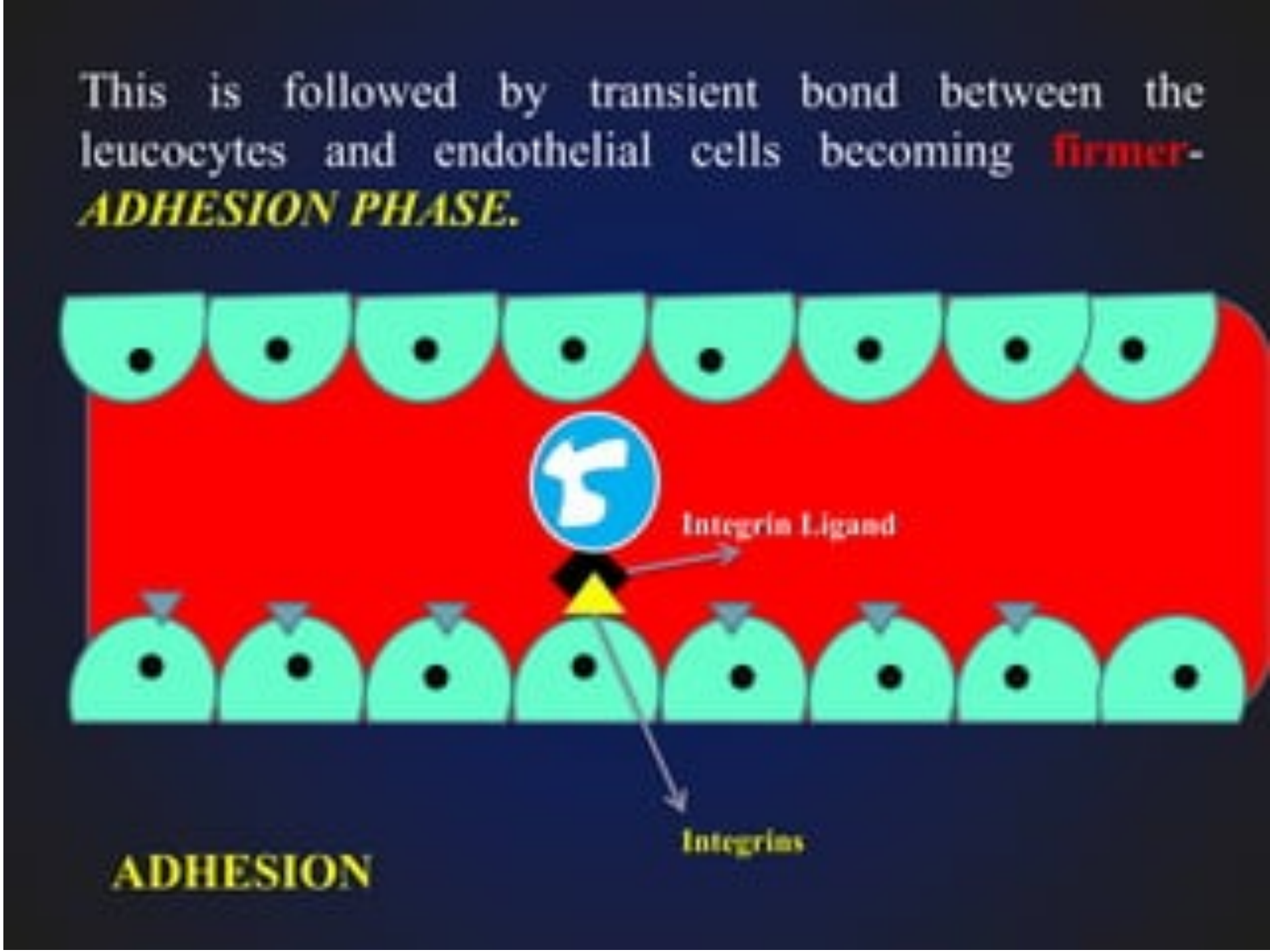


2nd step Rolling & Attachment (adhesion):

- *After margination of (PMN), leukocytes try to stick to endothelial wall*
- *There are two major families of adhesion molecules involved in leukocyte adhesion & migration , expressed on both leukocytes & endothelial cells.*
- *The expression is regulated by chemokines & cytokines (TNF & IL1)*
- *Leukocytes initially transiently adhere to vessel wall (loose attachment easily disrupted by blood flow) (bind, detach & bind again) ,process called (rolling) mediated by (selectins) molecule.*

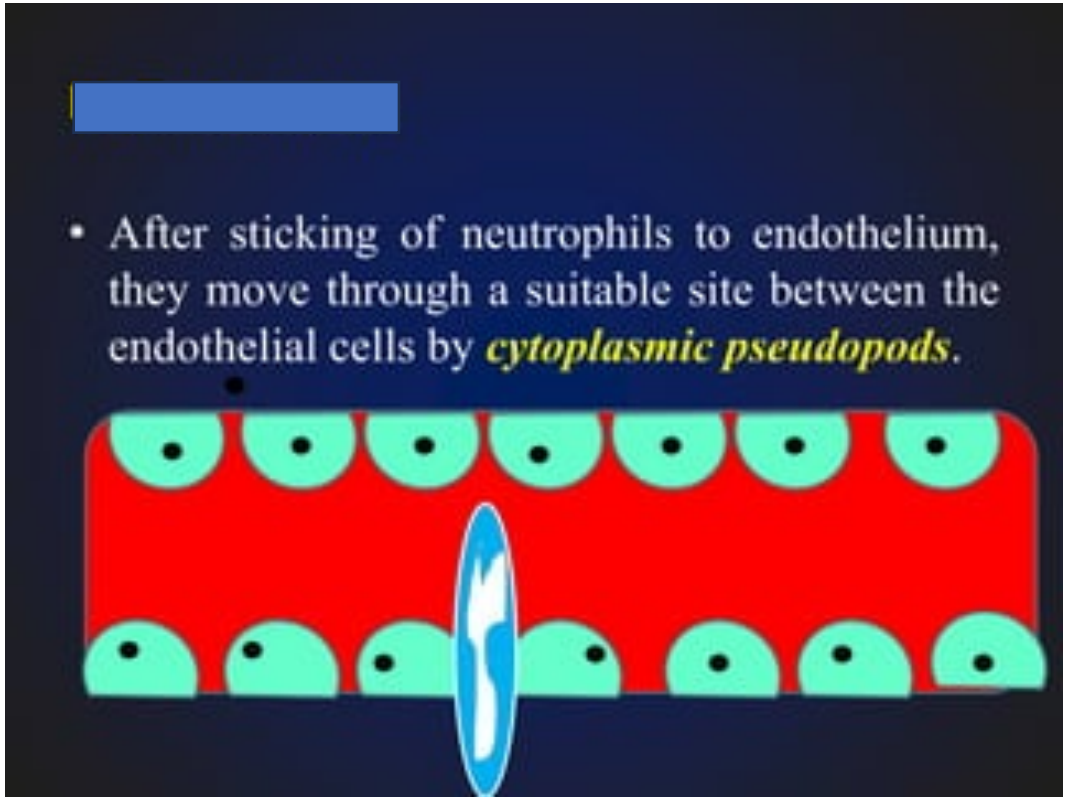


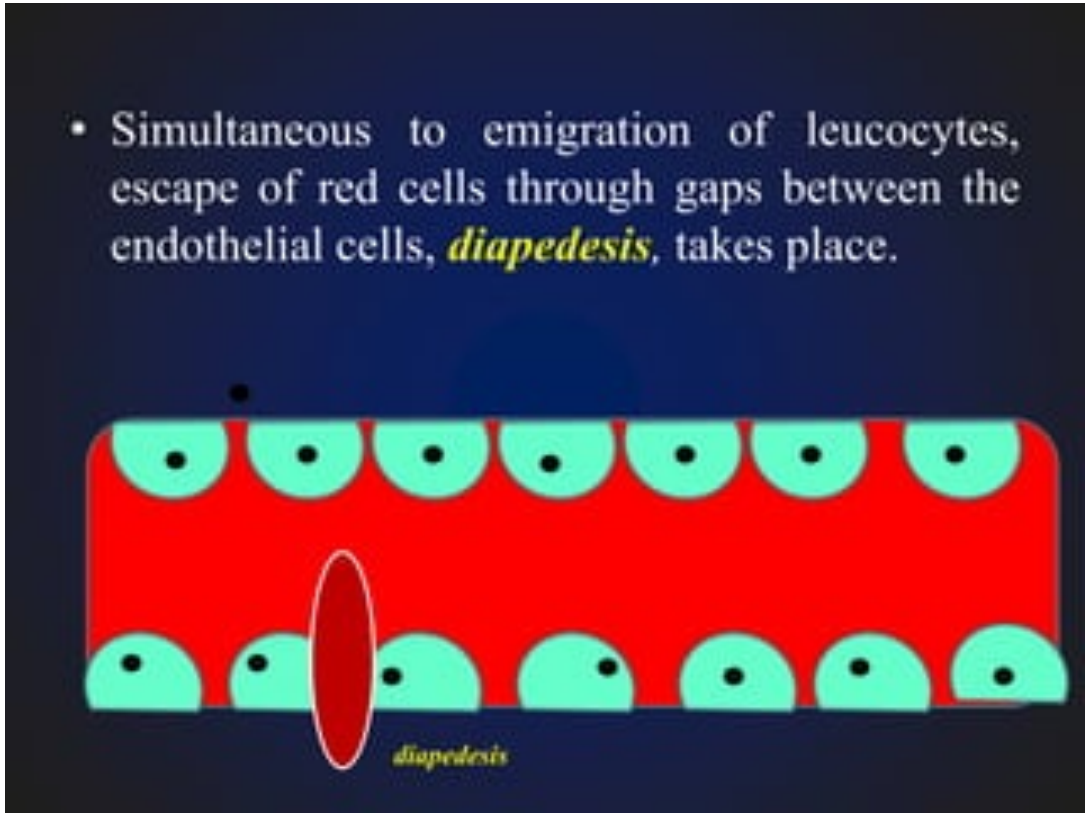
Then they adhere firmly to endothelial cells & is mediated by (integrins)



3rd step /Transmigration (Diapedesis)

PMNs is move through endothelial cell to EVS.





4th steps/ CHEMO-TAXIS :

The locomotion of leukocyte along the chemical gradient is called chemotaxis

What are the chemotactic stimuli ?

Both exogenous & endogenous substance can act as chemoattractant,

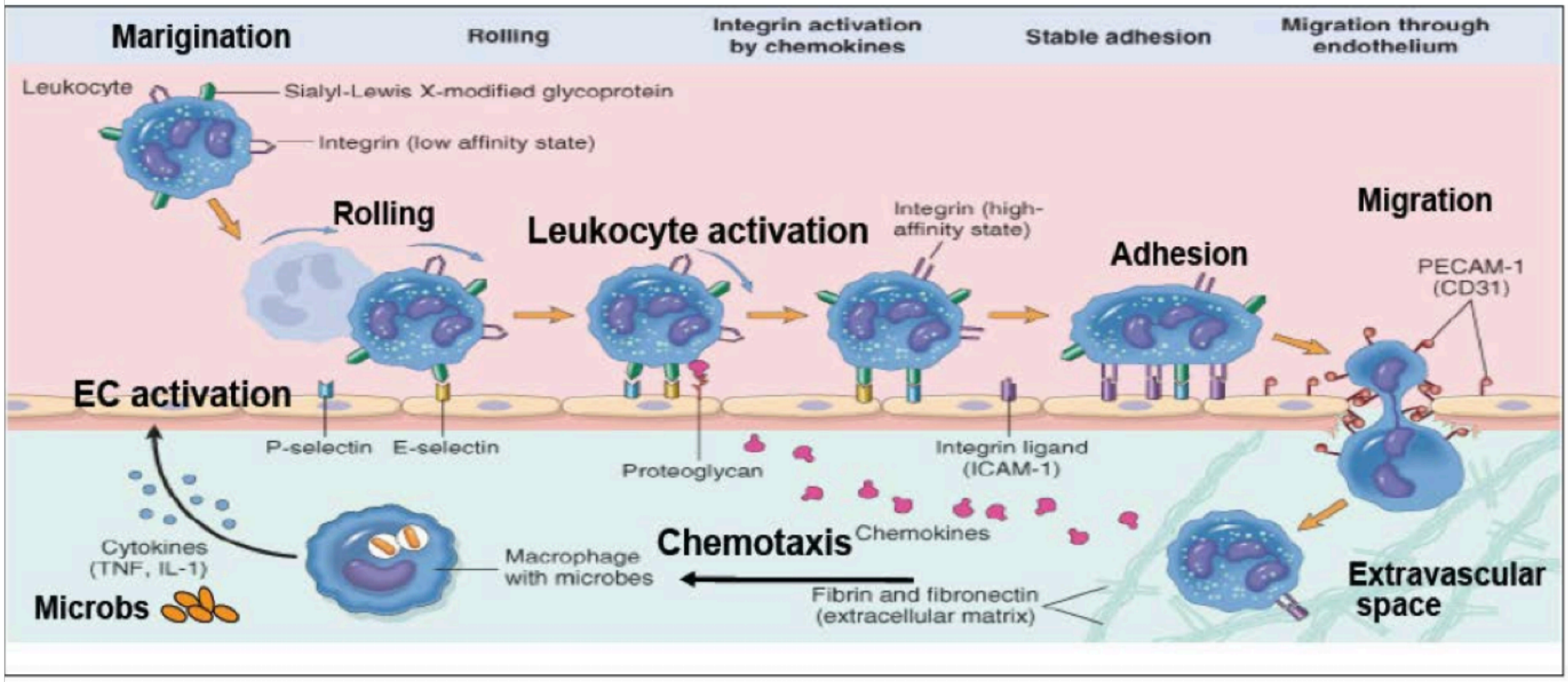
1- Bacterial products.

2- Cytokines. (Chemokine and TNF)

3- Components of the complement system, C5a

4- Products of the lipoxygenase pathway of arachidonic acid (AA) metabolism, particularly leukotriene B4 (LTB4) .

Leukocytes emigration (PMN predominate in early infl. infiltrate & later are replaced by monocytes & macrophage)

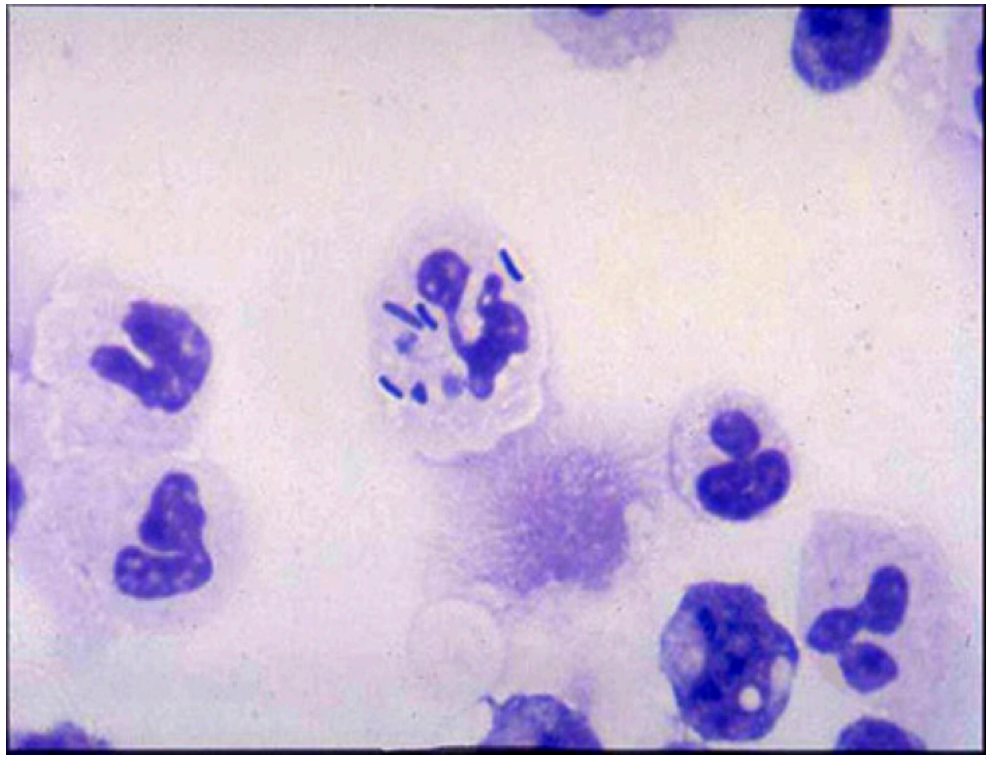


2. Phagocytosis.

- Phagocytosis is defined as the process of engulfment of solid particulate material by the cells (cell-eating).
- The cells performing this function are called ***phagocytes.***

Next step After leukocyte recuritment to EVS is process of *phagocytosis*

- *Three steps in correct order*
- *1-RECOGNITION & attachment, follow by*
- *2-ENGULFMENT & Phagocytosis, follow by*
- *3-KILLING & (DEGRADATION/ DIGESTION)*



1- Recognition & attachment

- *The Leukocytes to ingest micro organisms & dead cells, it must bind to specific protein called(Opsonins) that coat microbes & target them for phagocytosis , the process called opsonization.*
  The opsonin promote the adhesion between the particular material to be digested & the phagocyte's cell membrane.
  The leukocytes express receptor to the opsonin
  The Opsonins are :
  1- IgG antibody bind to microbial surface Ag.
  2-C3b Complement fragment

2. Engulfment.

Microbes recognized by receptors are ready to be engulfed by PMN/Macrophage

↓

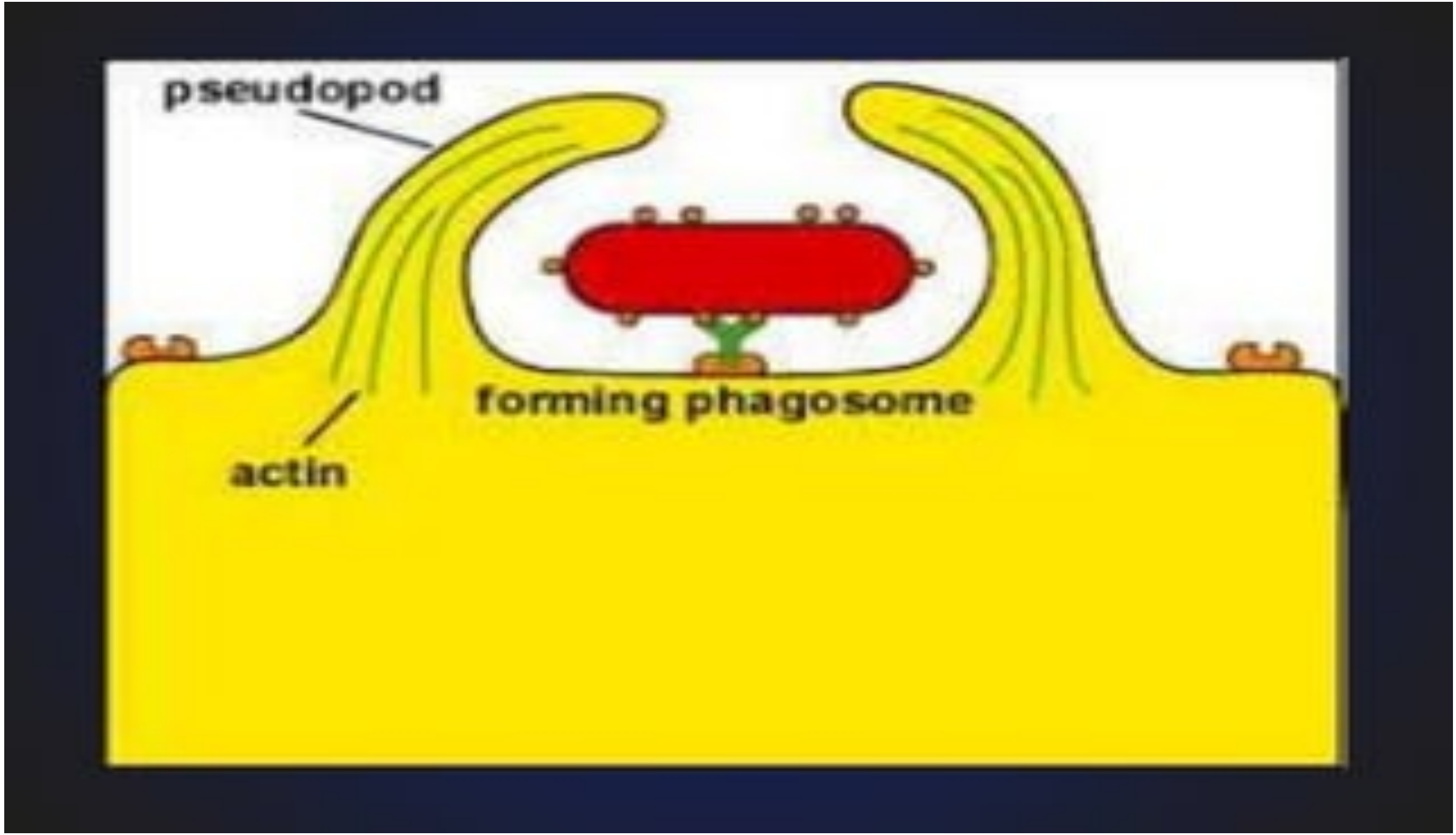
Cytoplasmic pseudopods of Phagocytes formed from activated actin filaments of cell wall will engulf the microbe

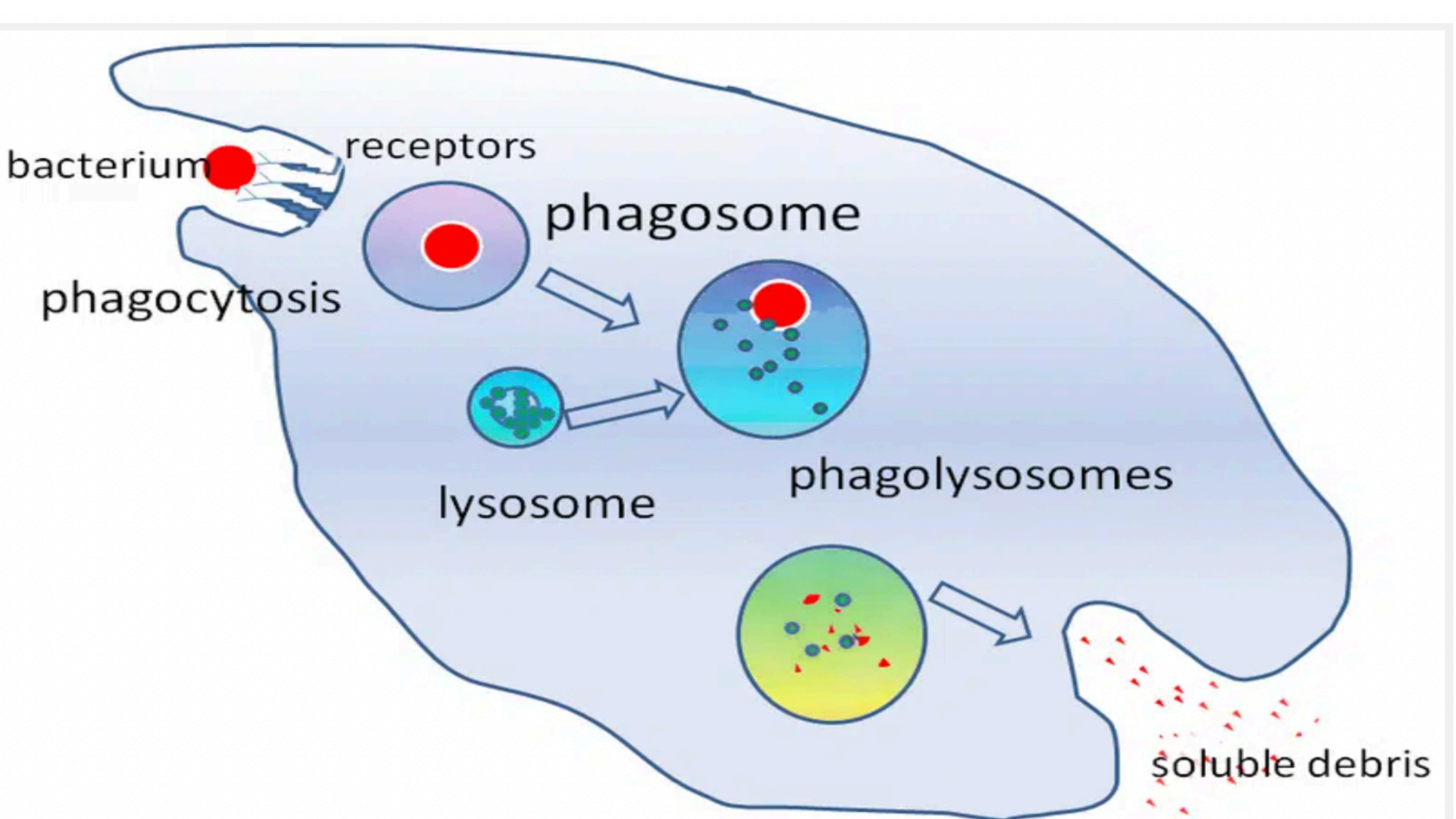
↓

Phagocytic vacuole (Phagosome) is formed in the cytoplasm from the detached cell membrane which previously engulfed microbe by its pseudopods.

↓

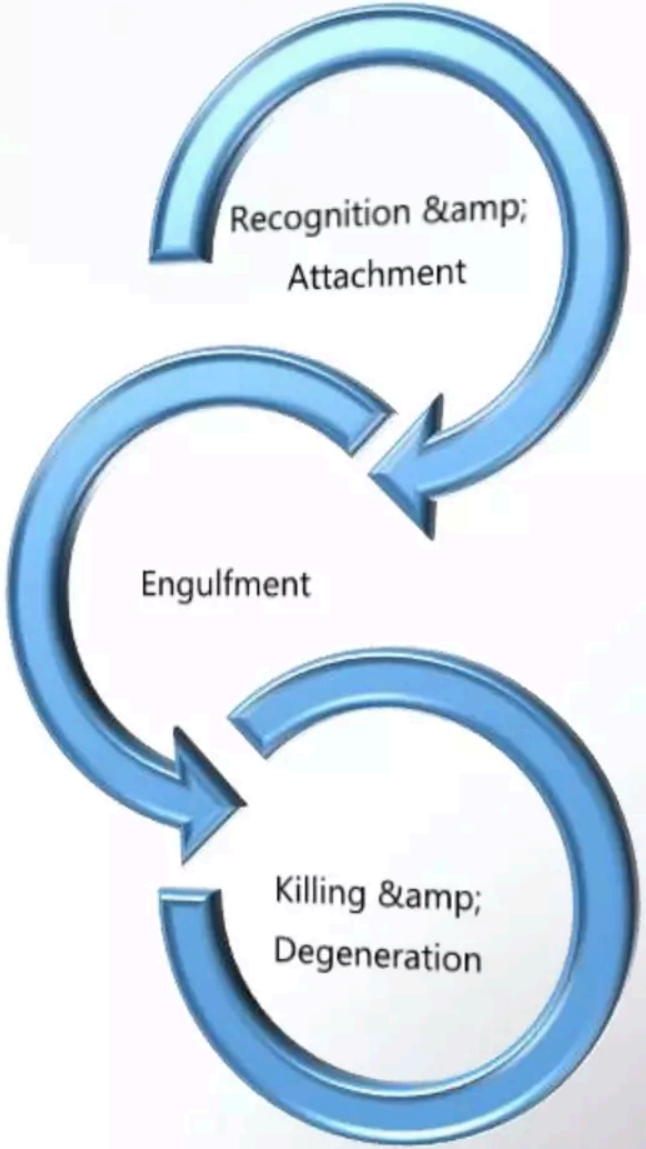
The phagosome fuses with one or more lysosomes of the cell and form bigger vacuole called phagolysosome.

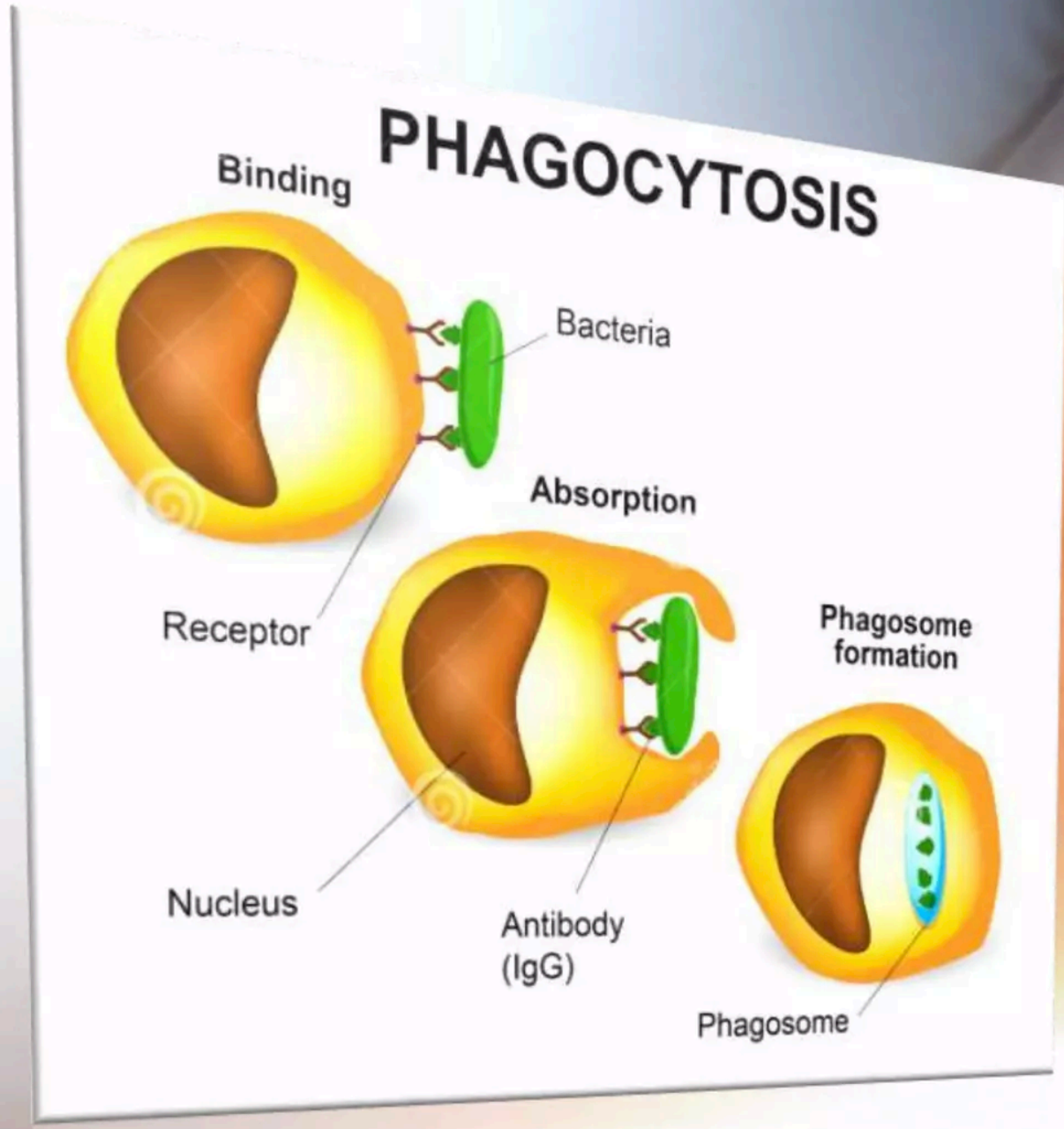

pseudopod
forming phagosome
actin


bacterium
receptors
phagocytosis
phagosome
lysosome
phagolysosomes
soluble debris

- During this process, lysosome contents may also leak into the extracellular space, resulting in tissue damage ("*Leukocyte-mediated tissue injury*) at the same time may contribute to host tissue damage (the harmful effect of inflammation)

SEQUENTIAL STEPS





Killing & Degradation process

- **Oxygen dependent mechanism (mostly).**
- **Oxygen independent mechanism**

❖ Oxygen-dependent mechanism (mostly)

- Two types of oxygen- dependent killing mechanisms

➢ **Myeloperoxidase–dependent(MPO)**

- **H_2O_2-Myeloperoxidase-halide system**: produces HOCl· (i.e. bleach!)

MPO-derived, reactive species is the most efficient bactericidal killing system in neutrophils. it is also effective against fungi, viruses, protozoa .

➢**Non-(MPO) dependent :** Uses superoxide **(O2) & (HO-** hydroxyl radicals.

- Less efficient.
- **killing of micro organism is due to formation of ROS such as hydrogen peroxide (H2O2) & other species have single unpaired electrons in their outer orbits that react with molecules in cell membrane or nucleus causing damages.**

killing in phagocytic vacuole (Oxygen dependent) mechanism

- *The free radicals generated within the phagolysosome, during process of phagocytosis*

- *- ROS are generated by the rapid activation of a phagocyte oxidase (NADPH oxidase) present on membrane of phagolysosome which convert oxygen to (O2) to superoxide ,this in turn converted to hydrogen peroxide - (H2O2) then to HOCL hypochlorite potent oxidant & anti microbial agent, generated by the action of* **MPO**

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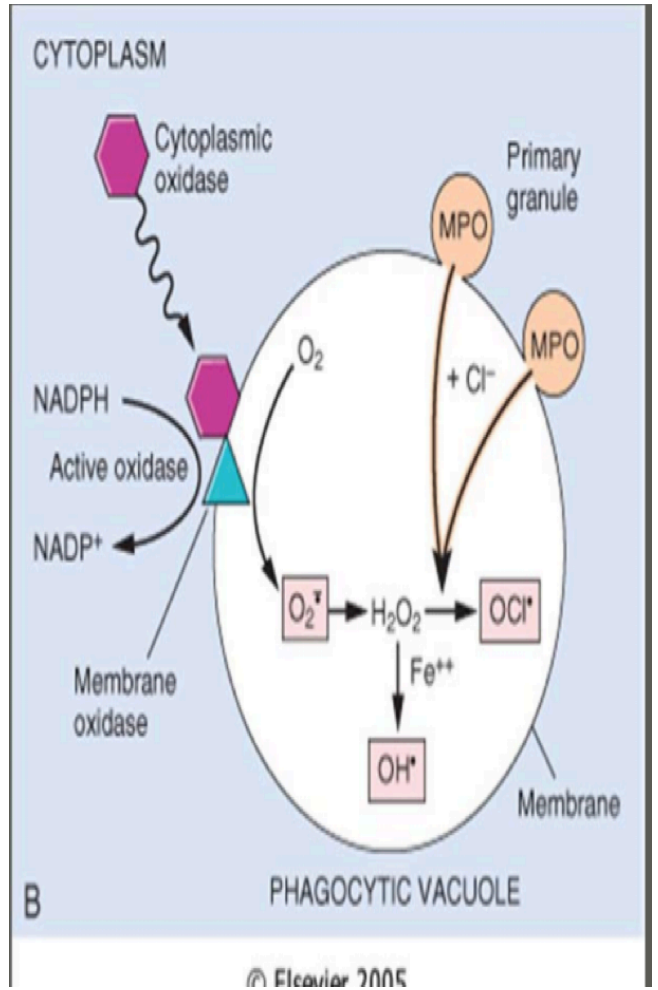


❖ Oxygen independent mechanism

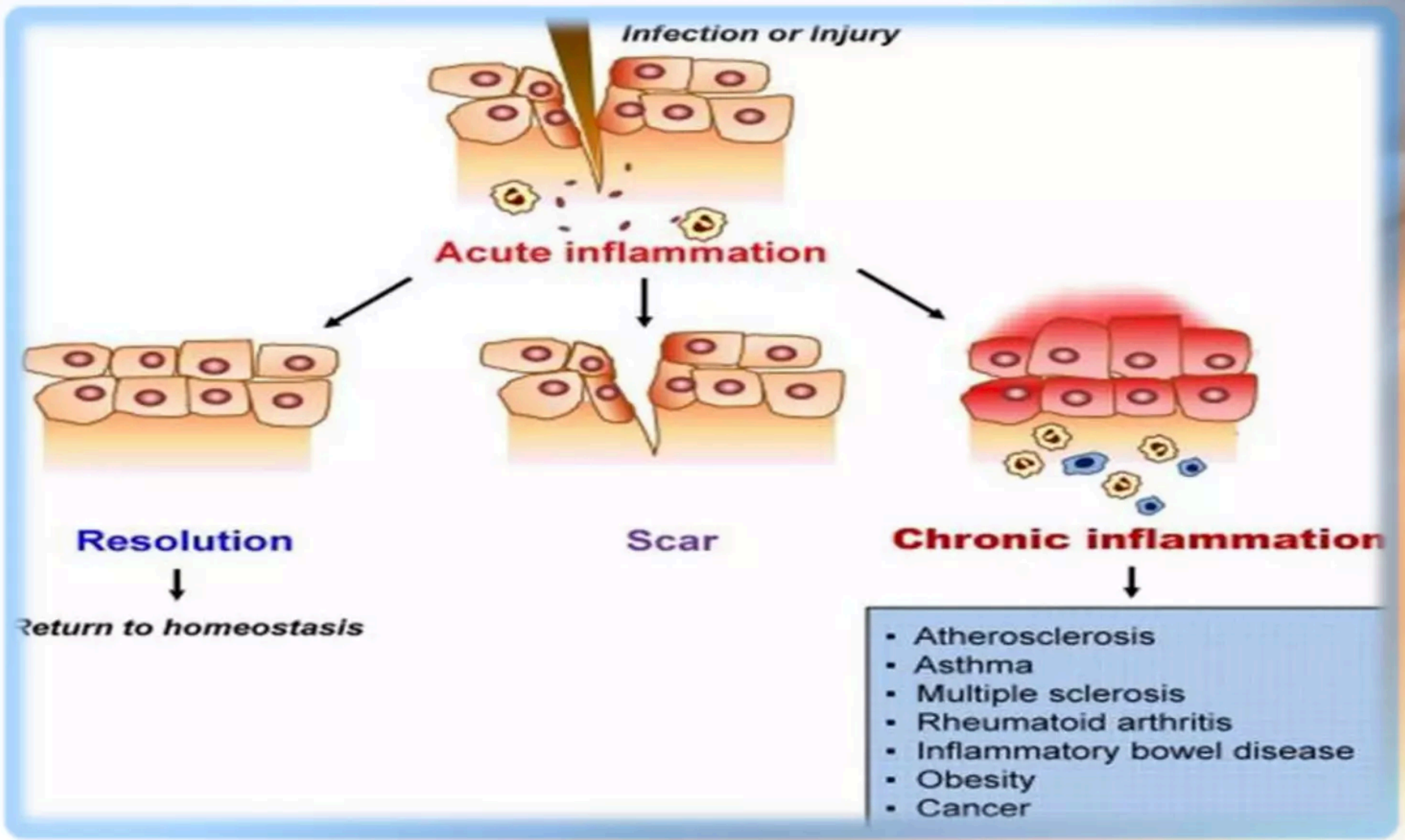
- **Oxygen independent mechanism :** in the lysosome Bactericidal permeability increasing protein (BPI) , Lysozymes, Lactoferrin, & Defenses,
- Major Basic Protein (MBP; Eosinophils)
- **Neutrophil Extracellular Traps (NETs)** formation, killing that does not involve phagocytosis , the extracellular microbes are killed by trapping micro organism in extracellular fibrillar networks formed to prevent the spread of the microbes .

Diseases with Defect in Leukocyte Function

- **Defect in leukocytes can be acquired & inherited – which increase susceptibility to inflammation.**
- **Acquired causes :**
- •**Bone marrow suppression** : caused by tumor or treatment with chemotherapy or radiation - result in decrease in leukocyte No.
- •**Metabolic disease** (diabetes)affect leukocyte function.
- **Inherited dis.(Genetic disorder)**
- Leukocyte adhesion deficiency -LAD**: 2 type** both result in impairment in adhesion , migration.
- **1-**(leukocytes adhesion deficiency type I: (LAD –I) defect in synthesis of **integrin.**
- **2-**leukocytes adhesion deficiency type 2 LAD-2 defect in **selectins**

- **-Defect in microbicidal activity ?.(CGD) in childhood -**
- **Chronic granulomatous disease** Defect in phagocyte oxidase enzyme. in **NADPH oxidase** responsible for generation of superoxide. Engulfment of bacteria does not result in activation of O2 dependent killing mechanism but more macrophage & neutrophils are recruited called to the site of inflammation. resulting in formation of granuloma formation .
- **- Defect in phagolysosome formation** disorder called **Chediak –Higashi syndrome** – results in recurrent pyogenic inf .

Fate of Acute Inflammation



Morphologic Patterns of Acute inflammation

Many variables may modify the basic inflammatory response;

these include:-

1. The nature and intensity of the injury.

2. The site and tissues affected.

3. The responsiveness of the host.

- ❑ *Several types of inflammation are recognized, which vary in their morphology and clinical correlates.*

Morphologic Patterns of Acute inflammation

- Serous inflammation
- Fibrinous inflammation
- Suppurative (purulent) inflammation -abscess formation
- Catarrhal inflammation, e.x common cold
- Membranous inflammation, e.x Corynebacterium diphtheria

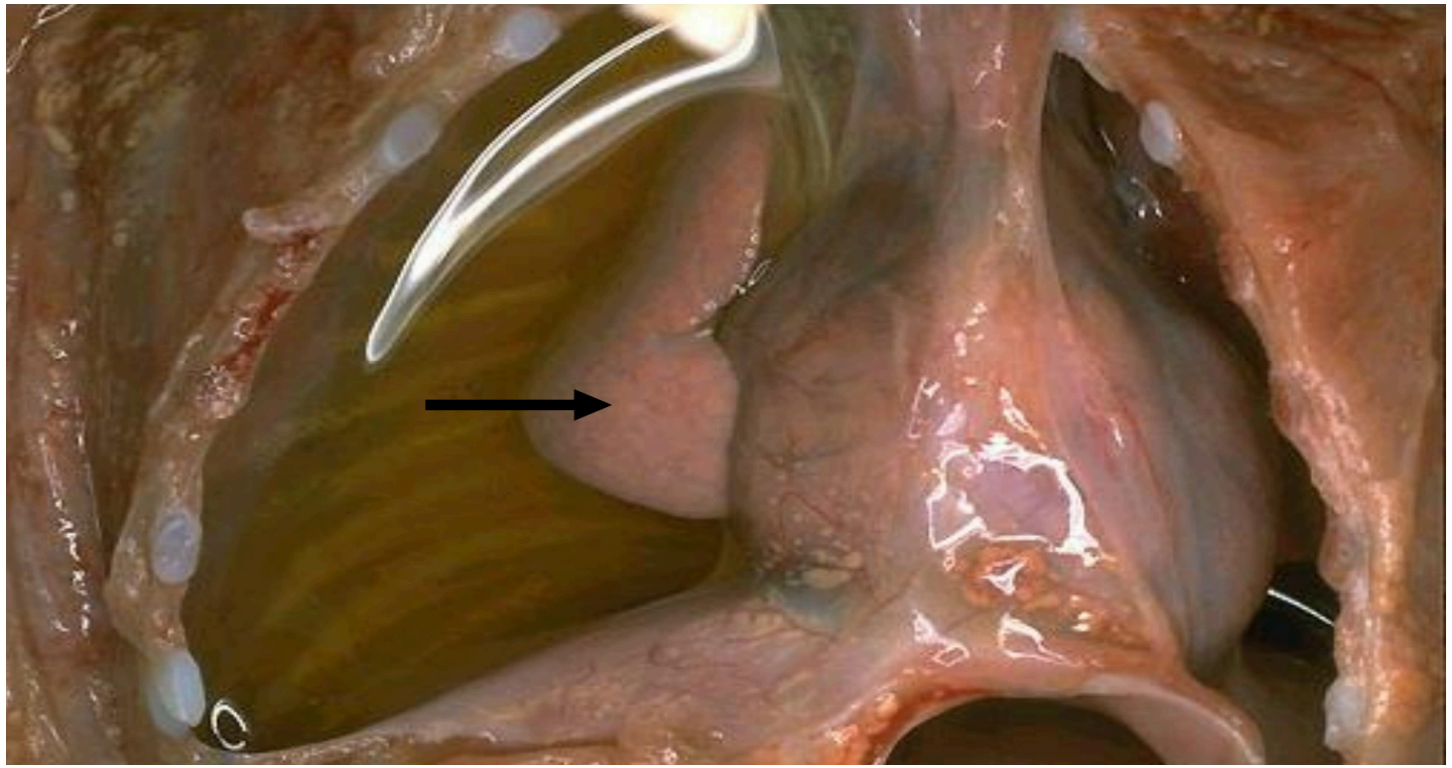
1- Serous inflammation

MORPHOLOGICAL TYPES OF ACUTE INFLAMMATION:

1. Serous Inflammation:

- Characterized by marked outpouring of a thin serous fluid.
- Serous exudate or effusion is yellow, straw-like in color and microscopically shows either few or no cells.
- **Example:** Pleural effusion as a complication of lobar pneumonia.

Serous inflammation (serous pleural effusion)



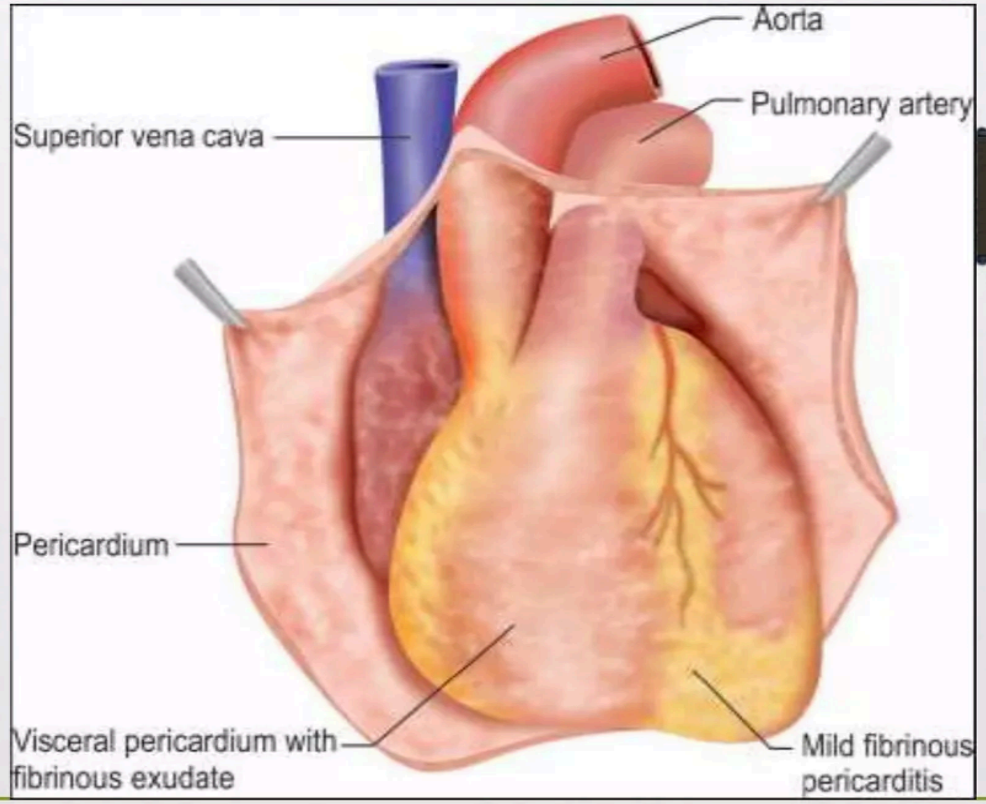
- *Excessive accumulation of clear, thin fluid within pleural cavity. It is transparent but note the reflection of light in the upper part of the photograph and lung collapse (arrow) due to pressure induced by the*

2-Fibrinous inflammation:

- Eosinophilic (pink) meshwork of threads / amorphous proteinaceous material that accumulates when vascular leaks are large enough for large molecules like fibrinogen to pass through the vessel wall, or there is local procoagulant stimulus (cancer).
- Usually seen in inflammation of lining of body cavity Serous cavities e.g. meninges, pericardium, pleura (SLE, uremia), or Mucosal surfaces e.g diphtheria .colon dysentery (Shigella), pseudomembranous colitis (Clostridium difficile)
- If not removed by fibrinolysis or macrophages, can organize into scar tissue and lead to fibrous thickening / obliteration of the body cavity space

MORPHOLOGICAL TYPES OF ACUTE INFLAMMATION:

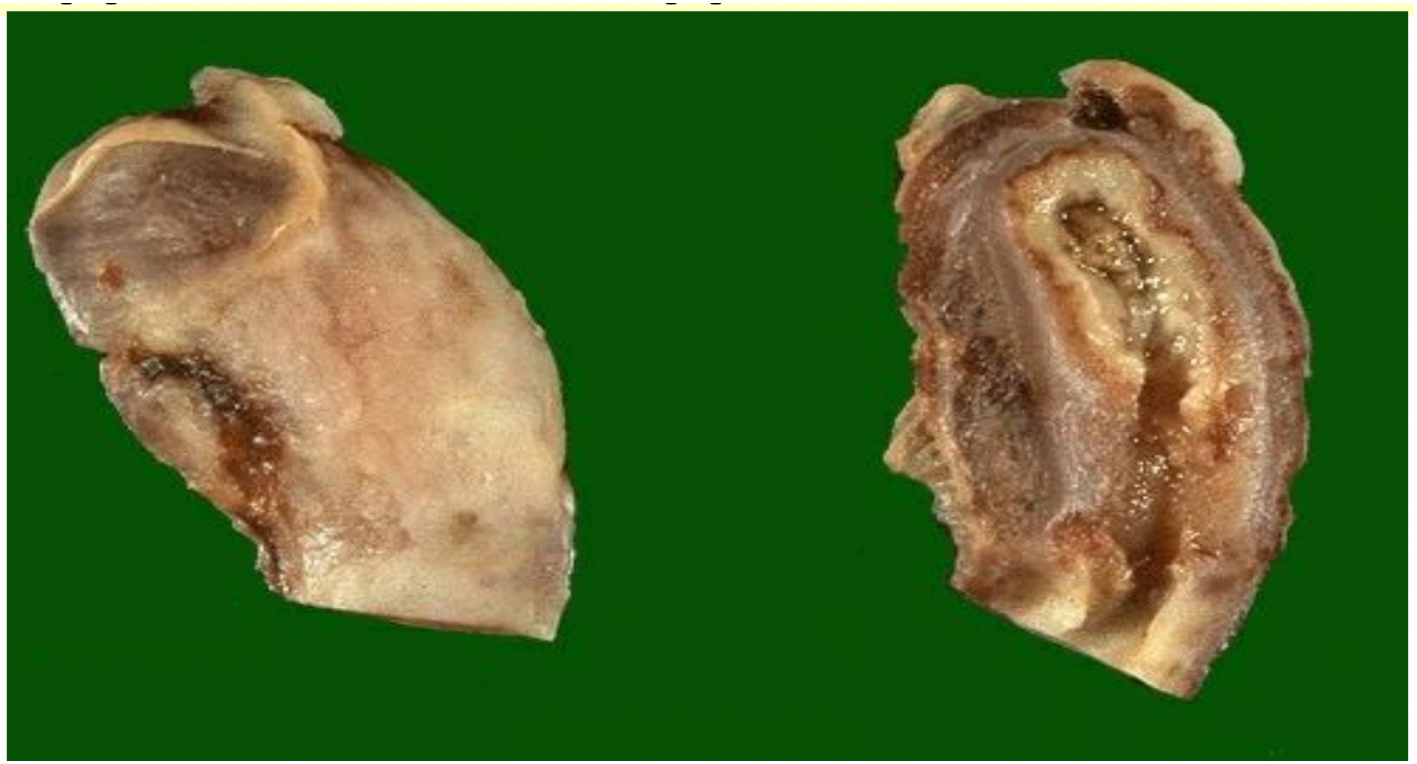
- The **exudate rich in fibrin** is called fibrinous exudate.
- For example, fibrinous pericarditis is seen in rheumatic fever and classically known as ***"bread and butter" pericarditis.***



3-Suppurative (purulent) inflammation

- **This is characterized by the production of large amounts of pus or purulent exudate consisting of neutrophils, necrotic cells, and edema fluid. Certain bacteria (e.g., staph. aureus, St. pyogenes, Pneumococci, gonococci, meningococci and E. coli) produce this localized suppuration and are therefore called pyogenic (pus-producing) bacteria. A common example of an acute suppurative inflammation is acute (suppurative) appendicitis.**

Appendix: acute suppurative inflammation



- *Upper half of excised appendix. Lt: fibrino -purulent serosal exudate Rt: lumen filled with pus*

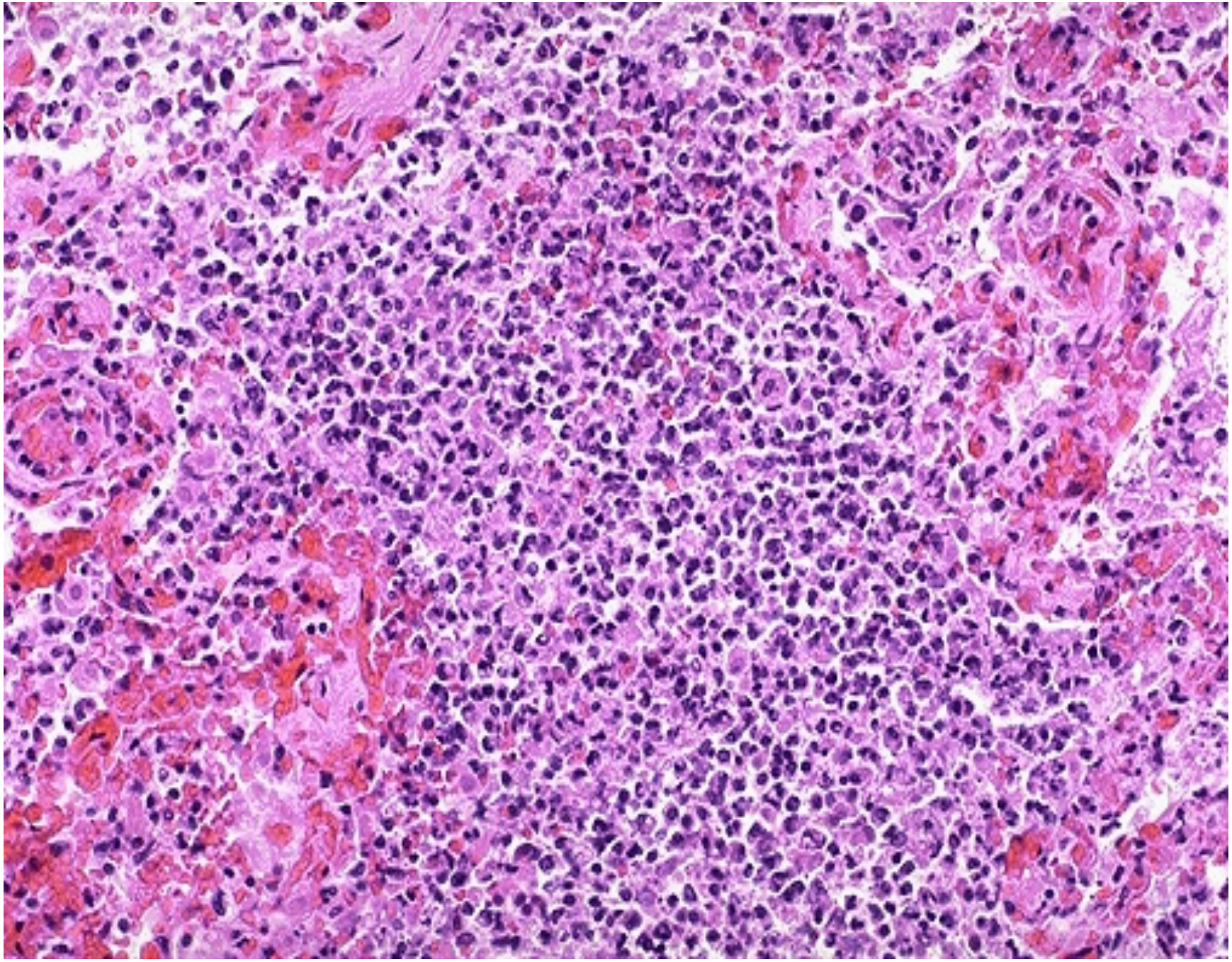
Abscess;

- *An **abscess** is a localized collection of purulent inflammatory fluid (pus) caused by suppuration buried in a tissue, an organ, or a confined space.*
- ***Pus** is a thick creamy yellow or blood-stained fluid.* **Abscesses** *are produced by deep seeding of pyogenic bacteria into a tissue. They have a central region that appears as a mass of necrotic leukocytes and tissue cells.*

Lung abscess

List the key histopathological features?

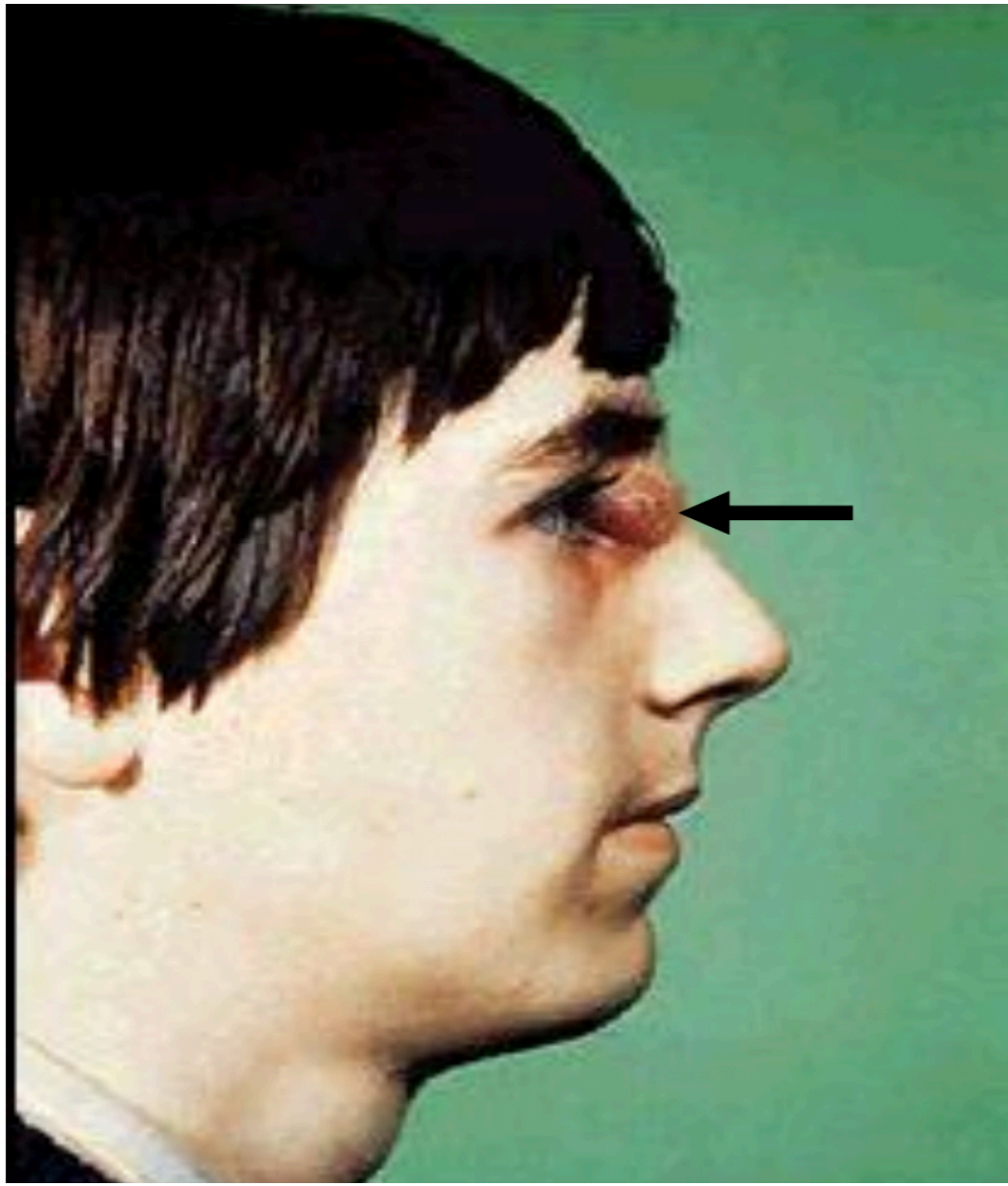
Suppurative destruction of lung parenchyma
Central area of cavitation filled with pus (neutrophils, necrotic issues,
Inflammatory exudates: fluid component



- *There is usually a zone of preserved neutrophils around this necrotic focus, and outside this region vascular dilation and fibroblastic proliferation occur, indicating the beginning of repair. In time, the abscess may become walled off and ultimately replaced by connective tissue. A common example of an abscess is the skin furuncle.*

Abscess (Furuncle) (boil)

- *Abscess that involves the skin is called “Boil” or “furuncle*



Ulcers :-

- *An ulcer is a local defect, or excavation of the surface of an organ or tissue that is produced by the sloughing (shedding) of inflammatory necrotic tissue. Ulceration occurs only when tissue necrosis and resultant inflammation exist on or near a surface.*

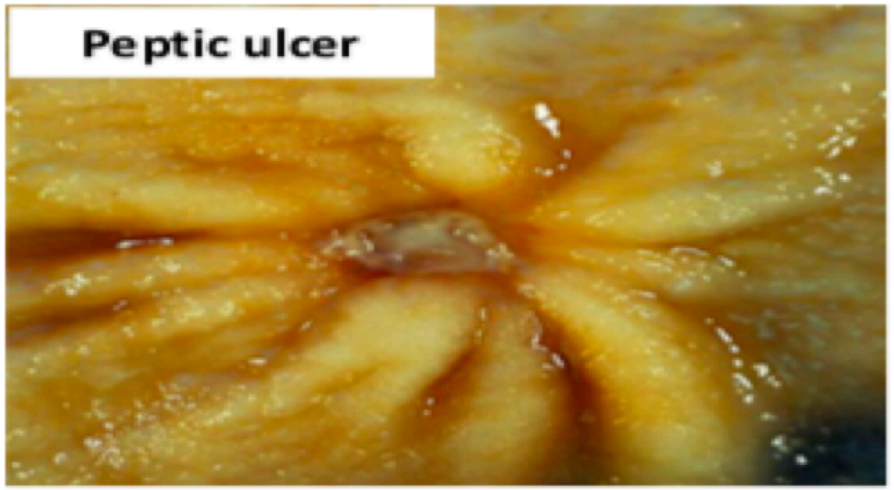
Ulcers, ulcerative inflammation

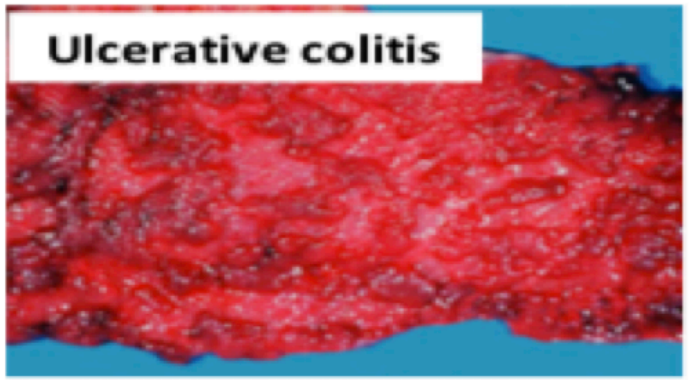
Basic features:

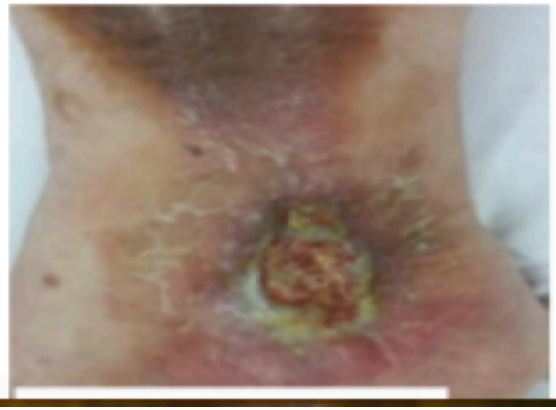
- NOT a type of inflammation, but frequently related to inflammation (can be both cause and consequence)
- Definition:
  - Defect of mucosa / skin AND underlying tissues caused by necrosis
  - Accompanied by inflammation, regeneration and scar formation

Examples:

- Peptic ulcer – stomach / duodenum
- Ulcerative colitis
- Venous congestion – skin of the lower extremities
- Ulcers of the oral cavity, genitourinary tract, etc.







SYSTEMIC EFFECTS OF ACUTE INFLAMMATION

- Systemic changes in acute inflammation are collectively known as **acute-phase response,** or the Systemic Inflammatory Response Syndrome (SIRS).
- **Causes:** Due to cytokines produced by leukocytes, in response to infections or immune reactions.
- Most important cytokines are TNF, IL-1, and IL-6.

Acute phase response

Symptom, physical exam or lab finding	Comments	Mediators
Fever, malaise, somnolence, anorexia		TNF IL-1 Prostaglandins (fever)
↑ Production of acute phase proteins (e.g. **fibrinogen** and CRP)	Nonspecific markers of inflammation (ESR)	IL-6
Leukocytosis	-Initial early release -Increased production of WBC in bone marrow	-TNF & IL-1 -Colony stimulating factors
Shock	Hypotension, DIC, acidosis	High levels of: TNF & IL-1

SYSTEMIC EFFECTS OF ACUTE INFLAMMATION

- **The Clinical and Pathologic changes of acute-phase response are:**

1. **Fever:**

- **Pyrogens:** These are molecules that cause fever. It may be **exogenous** (bacterial products, like **LPS**), which stimulate leukocytes to release **endogenous** pyrogens **(cytokines such as IL-1 and TNF)**.
- The cytokines increase the enzymes cyclooxygenases resulting in conversion of AA into **prostaglandins**.

SYSTEMIC EFFECTS OF ACUTE INFLAMMATION

2. Raised plasma levels of acute-phase proteins:

- These are plasma proteins synthesized in the liver and may be markedly raised in response to inflammatory stimuli.

Types of acute-phase proteins:

- **(1) C-reactive protein (CRP)**
- **(2) fibrinogen**
- **(3) serum amyloid A (SAA) protein.**

SYSTEMIC EFFECTS OF ACUTE INFLAMMATION

3. Changes in the leukocytes:

A. Leukocytosis

- Total leukocyte count more than 11,000/μL are termed as ***leukocytosis.***
- Common in inflammatory reactions, especially those caused by bacterial infections.

SYSTEMIC EFFECTS OF ACUTE INFLAMMATION

B. Lymphocytosis:

- It is seen in **viral infections** (e.g. Infectious mononucleosis, mumps, and German measles).

- **Erythrocyte Sedimentation Rate** Distance RBC fall in an hour, If there is an acute phase protein (IL-6 mediated fibrinogen)-->RBC stack (rouleaux)-->fall larger distance than normal cells (lowest bar)
- **Septic Shock** Severe infection, causes hypotension, dic, and metabolic disturbances, TNF and IL-1

SYSTEMIC EFFECTS OF ACUTE INFLAMMATION

Other features of the acute-phase response:

It includes:

- Increased pulse and blood pressure.
- Anorexia and malaise, probably due to cytokines acting on brain cells.
- In severe bacterial infections (sepsis) cytokines (mainly TNF and IL-1) may be produced in large quantities and can result in disseminated intravascular coagulation and cardiovascular failure.

Thank You